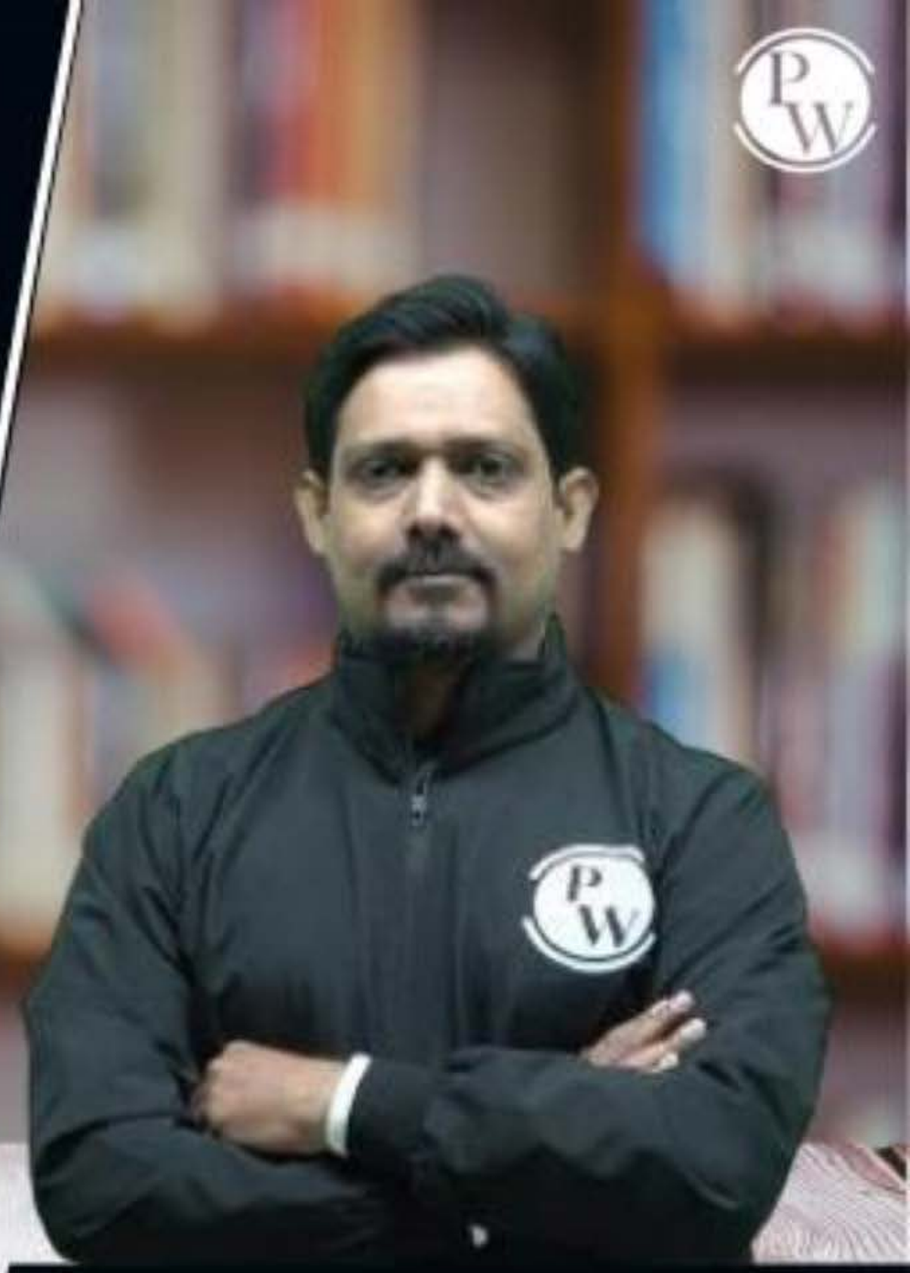
Computer Science & IT



Data Structure & Programming

Array

Lecture No. 04



By- Abhishek Sir

Recap of Previous Lecture



Topic 2-D Array Column Major Order

Topic Lower Triangular Matrix

Topic Upper Triangular Matrix

Topic

Topic

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Topics to be Covered



Topic Lower Triangular Matrix Column major Order

Topic Upper Triangular Matrix

Topic Tridiagonal matrix

Topic

Topic

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Topic: Lower Triagular Matrix

LTM (Square Matrix)

Rows = Columns

Lower Triangular matrix is a matrix in which element above principle diagonal are zero

a[1][1]	A[1][2] = 0	A[1][3] = 0
a[2][1]	a[2,2]	A[2][3] = 0
a[3][1]	a[3][2]	a[3][3]

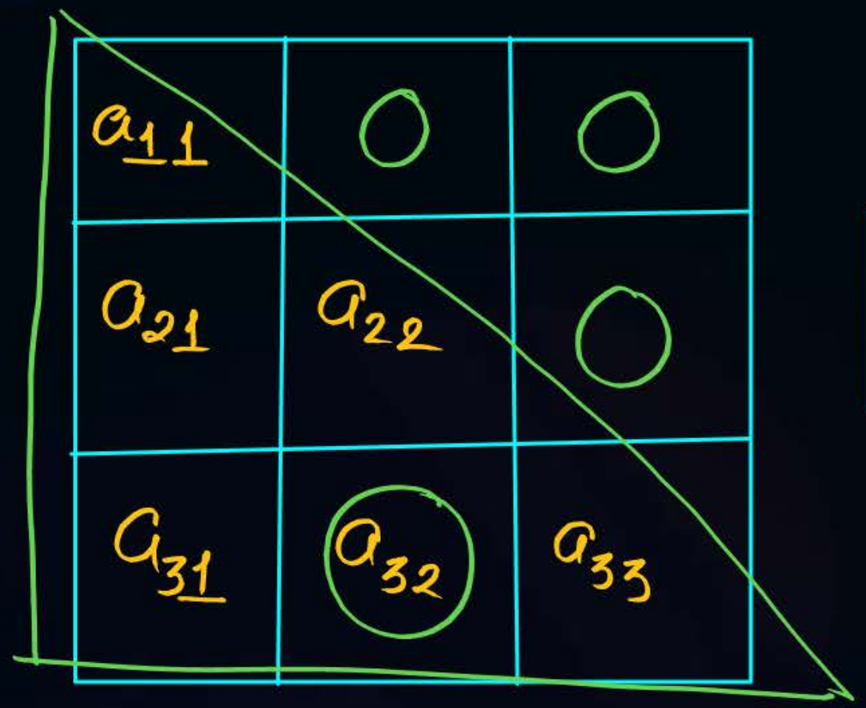
what is the condition on [i, j]

Such that A[i][j] = 0

j > i ⟹ A[i][j] = 0

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Question : Lower Triangular Matrix



Nonzero elements store

a_{11}	a_{21}	a_{22}	a_{31}	a_{32}	a_{33}	
1000	1002	1004	1006	1008	1010	

a[i][j] : Before i^{th} Row $(i-1)$ rows are Changed.

i^{th} Row :

j^{th} column : $j-1$ column Completed

No. of element : $1+2+\ldots+(i-1) = \frac{i(i-1)}{2}$

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Question : Lower Triangular Matrix

Row major order LTM :

$$BA + \left[\frac{i(i-1)}{2} + (j-1) \right] \times Size$$

address are unsigned int.

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Topic : Question

Address

Q A[1...100][1...100] is lower triangular matrix. Base address is 1000 and each element occupies 2-Bytes of space. What is the address of A[50][50] in row major order ?

A[50][50] — ?

A[10][20] — ?

$$A[50][50] = 1000 + \left[\frac{50 \times 49}{2} + 49\right] \times 2$$

Not stored.

$$= 1000 + [25 \times 49 + 49] \times 2$$

$$= 3548 \checkmark$$

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A[10][20]

↓ if(j>i)

return 0;

else

address calculate. $BA + \left[\frac{(i(i-1))}{2} + (j-1)\right] \times Size$

return value at that Address



Topic : Answer

Before reaching 50th row, number of rows completed is 49

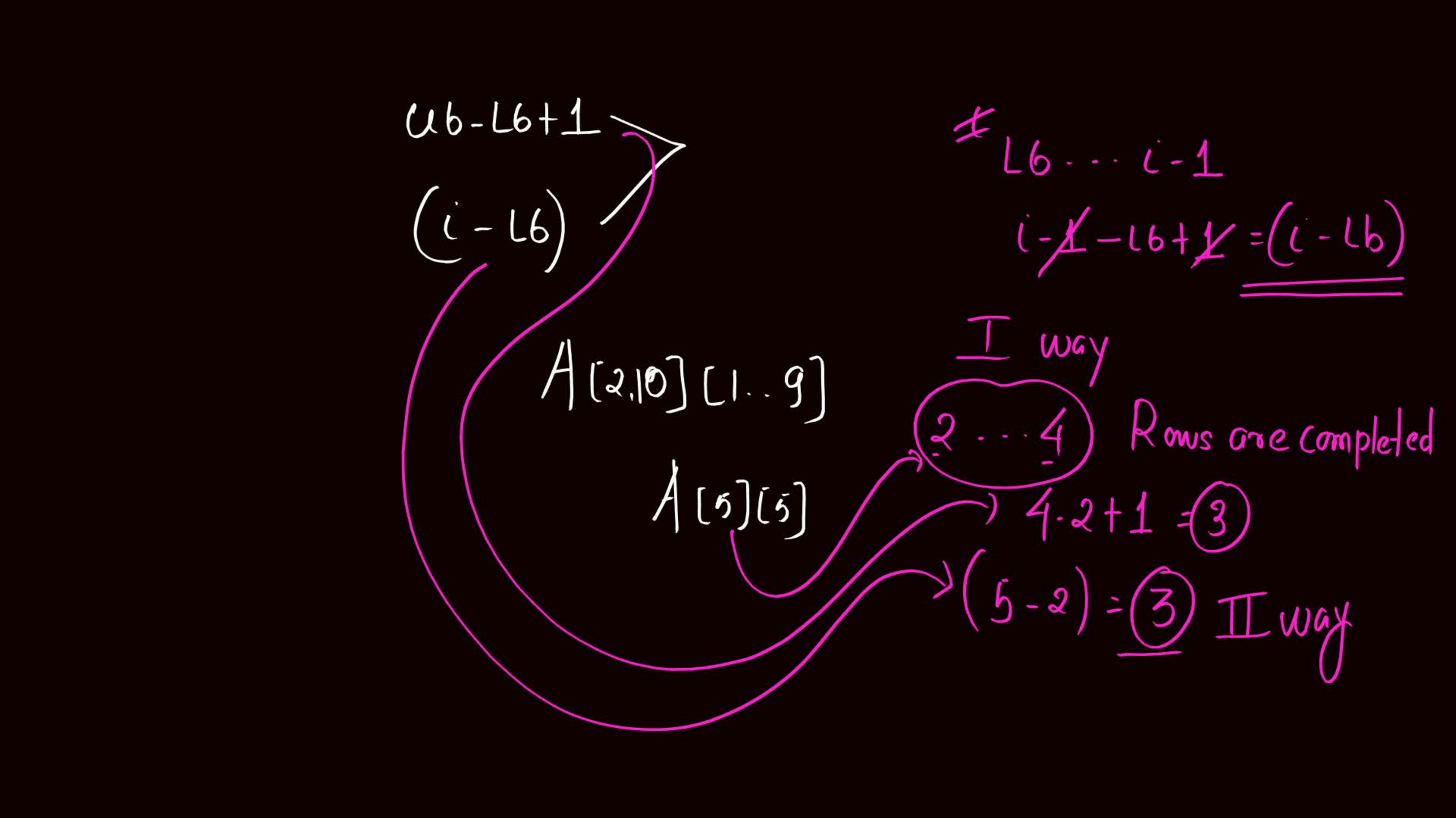
Total number of elements in 49 rows

1+2+3+…..+49 = (49×50)/2=1225

We are at 50th column , number of element completed is 49

Total number of element completed is 1225+49= 1274

Address of a[50][49] is = 1000+1274*2=3548


Ub-Lb+1
(i-Lb)
A[2,10][1..9]
A[5][5]
I way
2 ... 4
Rows are completed
4-2+1 = (3)
(5-2) = (3) II way
≠ Lb ... i-1
i-1-Lb+1 = (i-Lb)

Topic : Lower Triangular Matrix: Column Major Order

a_{11}	0	0	0
a_{21}	a_{22}	0	0
a_{31}	a_{32}	a_{33}	0
a_{41}	a_{42}	a_{43}	a_{44}

1000	1002	1004	1006	1008	1010	1012	1014	1016	1018
a_{11}	a_{21}	a_{31}	a_{41}	a_{22}	a_{32}	a_{42}	a_{33}	a_{43}	a_{44}

Column-1 | 3 elements | 2 element | 1 element

Slide

Topic : Lower Triangular Matrix: Column Major Order

a_{11}	0	0	0
a_{21}	a_{22}	0	0
a_{31}	a_{32}	a_{33}	0
a_{41}	a_{42}	a_{43}	a_{44}

1000	1002	1004	1006	1008	1010	1012	1014	1016	1018
a_{11}	a_{21}	a_{31}	a_{41}	a_{22}	a_{32}	a_{42}	a_{33}	a_{43}	a_{44}

Column-1 | 3 elements | 2 element | 1 element

A[i][j] Address

we are in jth column and j-1 column are Completed

Slide

Topic : Lower Triangular Matrix: Column Major Order

a_{11}	0	0	0
a_{21}	a_{22}	0	0
a_{31}	a_{32}	a_{33}	0
a_{41}	a_{42}	a_{43}	a_{44}

1000	1002	1004	1006	1008	1010	1012	1014	1016	1018
a_{11}	a_{21}	a_{31}	a_{41}	a_{22}	a_{32}	a_{42}	a_{33}	a_{43}	a_{44}

Column-1 — 3 elements — 2 element — 1 element

Column No — # of element

1st column : 4 (4×4) ✓

2nd column : 3

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Topic : Lower Triangular Matrix: Column Major Order

a_{11}	0	0	0
a_{21}	a_{22}	0	0
a_{31}	a_{32}	a_{33}	0
a_{41}	a_{42}	a_{43}	a_{44}

1000	1002	1004	1006	1008	1010	1012	1014	1016	1018
a_{11}	a_{21}	a_{31}	a_{41}	a_{22}	a_{32}	a_{42}	a_{33}	a_{43}	a_{44}

Column-1 — 3 elements — 2 element — 1 element

$n \times n$

$(J-1)^{th}$ column

1st column — $(n-0)$. one value lesser

2nd column — $(n-1)$

3rd column — $(n-2)$

⋮

$(J-1)^{th}$ Column — $(n-(j-2)) = n-j+2$

Slide

Topic : Lower Triangular Matrix: Column Major Order

a_{11}	0	0	0
a_{21}	a_{22}	0	0
a_{31}	a_{32}	a_{33}	0
a_{41}	a_{42}	a_{43}	a_{44}

1000	1002	1004	1006	1008	1010	1012	1014	1016	1018
a_{11}	a_{21}	a_{31}	a_{41}	a_{22}	a_{32}	a_{42}	a_{33}	a_{43}	a_{44}

Column-1 — 3 elements — 2 element — 1 element

$n \times n$

1st column — (n)

+

2nd column — $(n-1)$

+

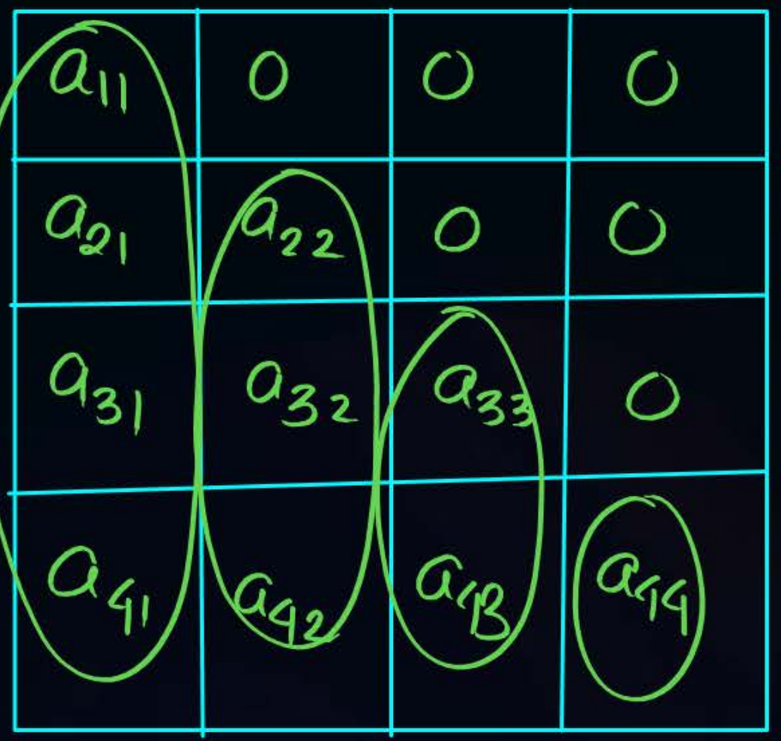
3rd column — $(n-2)$

$\vdots$

$(J-1)^{th}$ Column — $(n-(j-2))$

Slide

Topic : Lower Triangular Matrix: Column Major Order



1000	1002	1004	1006	1008	1010	1012	1014	1016	1018
a_{11}	a_{21}	a_{31}	a_{41}	a_{22}	a_{32}	a_{42}	a_{33}	a_{43}	a_{44}

Column-1 | 3 elements | 2 element | 1 element

$$n + n-1 + n-2 + \cdots + n-(j-2) \quad (j-1 \text{ terms})$$

$$\boxed{n(j-1)} - 1 - 2 - 3 \cdots - (j-2)$$

$$n(j-1) - (1+2+3 \cdots + j-2)$$

$$\boxed{\frac{n(n+1)}{2}} \quad n = j-1$$

$$\frac{(j-2)(j-1)}{2}$$

$$\boxed{n(j-1) - \frac{(j-1)(j-2)}{2}}$$

Slide

Topic : Lower Triangular Matrix: Column Major Order

a_{11}	0	0	0
a_{21}	a_{22}	0	0
a_{31}	a_{32}	a_{33}	0
a_{41}	a_{42}	a_{43}	a_{44}

1000	1002	1004	1006	1008	1010	1012	1014	1016	1018
a_{11}	a_{21}	a_{31}	a_{41}	a_{22}	a_{32}	a_{42}	a_{33}	a_{43}	a_{44}

Column-1 | 3 elements | 2 element | 1 element

No. of elements arranged in $(j-1)$ column

j^{th} column : $A[i][j] = \boxed{(i-j)}$ elements are stored.

$$\text{Address of } A[i][j] = \boxed{BAt\left[n(j-1) - \frac{(j-1)(j-2)}{2} + (i-j)\right] \times \text{Size}}$$

Slide



Topic : Lower Triangular Matrix: Column Major Order

a_{11}	0	0	0
a_{21}	a_{22}	0	0
a_{31}	a_{32}	a_{33}	0
a_{41}	a_{42}	a_{43}	a_{44}

1000	1002	1004	1006	1008	1010	1012	1014	1016	1018
a_{11}	a_{21}	a_{31}	a_{41}	a_{22}	a_{32}	a_{42}	a_{33}	a_{43}	a_{44}

Column-1 | 3 elements | 2 element | 1 element

Slide

Topic : Lower Triangular Matrix: Column Major Order

a_{11}	0	0	0
a_{21}	a_{22}	0	0
a_{31}	a_{32}	a_{33}	0
a_{41}	a_{42}	a_{43}	a_{44}

4x4

1000	1002	1004	1006	1008	1010	1012	1014	1016	1018
a_{11}	a_{21}	a_{31}	a_{41}	a_{22}	a_{32}	a_{42}	a_{33}	a_{43}	a_{44}

Column-1 | 3 elements | 2 element | 1 element

4x4

A[4][3] L = 4, J = 3, n = 4

$$1000 + \left[4 \times 2 - \frac{2 \times 1}{2} + 1\right] \times Size$$

$$1000 + [8 - 1 + 1] \times 2 = 1000 + 8 \times 2 = 1016$$

Slide



Topic : Lower Triangular Matrix: Column Major Order

LTM Row major order : $BA + \left[\frac{i(i-1)}{2} + j - 1\right] \times Size$

LTM Column major order : $BA + \left[n(j-1) - \frac{(j-1)(j-2)}{2} + (i-j)\right] \times Size$

Homework

UTM Row major order

UTM Column major order

$$\begin{matrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{matrix}$$

Slide



Topic :

Q

Suppose one LTM a [1..10][1..10] is given what is the Address of A[5][5] in Rowmajor order 2 in column major order. BA = 1000. Size = 2B

Row major order

$$1000 + \left[\frac{5 \times \overset{2}{4}}{\underset{2}{}} + 4\right] \times 2$$

$$1000 + 14 \times 2 = 1028$$

Slide



Topic :

Q

$n = 10$ $n \times n = 10 \times 10$

Suppose one LTM a [1..10][1..10] is given what is the Address of A[5][5] in Rowmajor order 2 in column major order. BA = 1000. Size = 2B

Column major order

$$1000 + \left[10 \times 4 - \frac{4 \times 3}{2} + 0\right] \times 2 = 1000 + [40 - 6] \times 2$$

$1000 + 34 \times 2 = 1068$

Slide



Topic :

Suppose 2 Dimensional array is given where each Row consist of 4 elements an element is stored at 1084. if element is accessed by A[i][j] then what is the value of (i+j) (Assume BA is 1000 Size = 2B. Lower Bound starts from 1)

(RMO)

Slide



To

(1000 ——— 1083) (1084) A[11][3] 1085

= 84B

No. of element = $\frac{84B}{2B}$ = 42

A[11][3]

= 1000 + [(11-1)×4 + (3-1)] × 2

1000 + [42] × 2

1084



Slide



Topic : Lower Triangular Matrix: Column Major Order

$$1084 = 1000 + \underbrace{((i-1)\times 4 + (j-1))}_{42} \times 2$$

$$(i-1)\times 4 + (j-1)$$

$$1000 + \underbrace{[(i-1)\times 4 + (j-1)]}_{40 \quad +2 \quad 42} \times 2 = 1084$$

$$1000 \quad [(11-1)\times 4 + (3-1)] \times 2$$

$i = 10$, $j = 70$

4 Rows
j can't be
(7)

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2 mins Summary

Topic — LTM Column major

Topic — practice problem

Topic

Topic

Topic

THANK - YOU